PROJECT REPORT

Customer Churn Prediction Using Machine Learning & Business Intelligence

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1. Dataset Overview

1.1 Dataset Description

This project analyzes customer churn in the telecommunications industry using historical customer data. The dataset contains demographic information, account characteristics, customer behavior, revenue-related attributes, and service usage statistics. Each record represents an individual customer, enabling both predictive modeling and business intelligence reporting.

The dataset includes a mixture of categorical and numerical variables, making it suitable for exploratory data analysis, feature engineering, machine learning classification, and interactive dashboard development.

1.2 Business Domain

The project belongs to the telecommunications sector, where customer retention is a major business objective. Acquiring new customers is generally more expensive than retaining existing ones, making churn prediction a valuable business application.

1.3 Business Problem

Customer churn directly impacts revenue and profitability. The objective of this project is to identify customers who are likely to discontinue their services so that the company can proactively implement retention strategies.

1.4 Target Variable

The target variable is **Churn**, where:

- **0:** Customer retained
- **1:** Customer churned

This represents a binary classification problem.

1.5 Project Objectives

The primary objectives are:

- Understand customer behavior.
- Identify factors associated with churn.
- Prepare data for predictive modeling.
- Develop multiple machine learning models.
- Compare model performance.
- Build an interactive Power BI dashboard.
- Provide business recommendations.

2. Data Preparation

2.1 Data Quality Issues

The original dataset contained several challenges that required preprocessing before model development:

- Missing values in selected features.
- Mixed numerical and categorical variables.
- Variables requiring encoding.
- Potential outliers in revenue and usage-related columns.
- Features measured on different scales.

2.2 Data Cleaning

The following preprocessing steps were performed:

- Removed unnecessary columns.
- Handled missing values using appropriate strategies.
- Checked for duplicate records.
- Converted data types where required.
- Standardized categorical values.

These steps improved data consistency and model reliability.

2.3 Feature Engineering

Several features were prepared to improve predictive performance:

- Label encoding for categorical variables.
- Feature scaling for numerical variables.
- Creation of machine-learning-ready datasets.

The final feature matrix contained only variables suitable for supervised learning.

2.4 AI-Assisted Preprocessing

Artificial intelligence tools were used to assist in:

- Identifying preprocessing strategies.
- Suggesting feature engineering approaches.
- Reviewing data quality issues.
- Accelerating exploratory analysis.

All AI-generated suggestions were manually reviewed before implementation.

3. Exploratory Data Analysis

EDA was conducted to understand customer behavior and identify patterns related to churn.

Key Findings

- **Customer Distribution:** The dataset contains both retained and churned customers, allowing supervised classification.
- **Revenue Analysis:** Average customer revenue varies across different customer groups, suggesting that financial characteristics influence churn behavior.
- **Geographic Analysis:** Customer churn differs across geographic areas, indicating that location may affect customer retention.
- **Demographic Analysis:** Variables such as marital status and income level demonstrate varying relationships with customer churn.
- **Correlation Analysis:** Correlation analysis identified relationships among numerical variables and highlighted features with greater predictive importance.
- **Outlier Detection:** Revenue and usage variables exhibited several extreme observations. These were reviewed during preprocessing to reduce their impact on predictive models.

4. Machine Learning

Models Implemented

To predict customer churn, three supervised classification algorithms were trained and evaluated:

- **Logistic Regression:** Logistic Regression was used as the baseline model because it is simple, fast, and highly interpretable. It establishes a reference point for comparing more advanced algorithms.
- **Random Forest:** Random Forest is an ensemble learning algorithm that builds multiple decision trees and combines their predictions. It is capable of capturing nonlinear relationships and interactions between customer features.
- **XGBoost:** Extreme Gradient Boosting (XGBoost) is an advanced boosting algorithm that sequentially improves weak learners by minimizing prediction errors. It is widely recognized for its high predictive performance on structured datasets.

Evaluation Metrics

The models were evaluated using five classification metrics:

- **Accuracy:** Overall percentage of correct predictions.
- **Precision:** Percentage of predicted churners that actually churned.
- **Recall:** Percentage of actual churners correctly identified.
- **F1-Score:** Harmonic mean of Precision and Recall.
- **ROC-AUC:** Measures the model's ability to distinguish churned from retained customers across different classification thresholds. This was selected as the primary evaluation metric because customer churn prediction requires balancing true positive and false positive rates.

Model Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.5923	0.5911	0.5760	0.5834	0.6295
Random Forest	0.6223	0.6085	0.6672	0.6365	0.6773
XGBoost	0.6393	0.6321	0.6513	0.6415	0.6957

Best Model by Metric

Metric	Best Model	Score
Accuracy	XGBoost	0.6393
Precision	XGBoost	0.6321
Recall	Random Forest	0.6672
F1-Score	XGBoost	0.6415
ROC-AUC	XGBoost	0.6957

Best-Performing Model & Justification

Based on the evaluation results, XGBoost achieved the strongest overall performance, obtaining the highest Accuracy (63.93%), Precision (63.21%), F1-Score (64.15%), and ROC-AUC (69.57%). Although Random Forest produced a slightly higher Recall (66.72%), XGBoost provided a better overall balance between identifying churned customers and minimizing false predictions.

The project selected XGBoost as the final predictive model because:

- It achieved the highest ROC-AUC (0.6957), the primary evaluation metric.
- It produced the highest Accuracy, Precision, and F1-Score among all tested models.
- It demonstrated a balanced trade-off between Recall and Precision, making it more reliable for customer churn prediction.
- The model's ability to capture complex nonlinear relationships makes it suitable for customer behavior analytics.
- It reduced false negatives and generalized effectively to unseen customer data.

For these reasons, XGBoost was integrated into the final business intelligence workflow, where its predictions support customer retention strategies and identify high-risk customer segments for targeted intervention.

5. Dashboard Summary

The Power BI dashboard consists of three interactive pages designed for business stakeholders:

- **Page 1 — Executive Overview:** Provides executives with a high-level overview of customer churn. KPIs include Total Customers, Churn Rate, Revenue Metrics, and Customer Distribution, allowing management to quickly assess overall business performance.
- **Page 2 — Customer Segmentation:** Identifies customer groups most associated with churn. Visualizations include geographic churn analysis, income distribution, marital status segmentation, and credit card ownership comparison to help pinpoint high-risk segments.
- **Page 3 — Predictive Insights:** Explains the factors driving customer churn using revenue KPI cards, Key Influencers, a Decomposition Tree, and executive recommendations to support strategic decision-making through AI-powered insights.

6. KPI Explanation

- **Total Customers:** Total number of customers in the dataset. Target: Maintain or increase customer base. Justification: A growing customer base supports long-term business growth. Performance: Indicates the scale of customers included in the analysis.
- **Churn Rate:** Percentage of customers who discontinued the service. Target: Minimize churn percentage. Justification: Lower churn increases customer lifetime value. Performance: Current churn rate highlights the need for targeted retention strategies.
- **Revenue Retained:** Average revenue generated from retained customers. Target: Maximize retained customer revenue. Justification: Retaining high-value customers improves profitability.
- **Revenue Lost:** Average revenue associated with churned customers. Target: Minimize revenue loss. Justification: Reducing churn protects recurring revenue.
- **Revenue Difference:** Difference between retained and churned customer revenue. Target: Increase the positive gap. Justification: Demonstrates the financial value of customer retention.

7. AI-Generated Insights

AI-powered Power BI visuals provided additional analytical capabilities:

- **Key Influencers:** Identified the customer characteristics most strongly associated with churn, enabling targeted retention campaigns based on customer behavior and demographics.
- **Decomposition Tree:** Allowed interactive exploration of churn across multiple dimensions such as geographic area, income, and customer characteristics.
- **Business Recommendations:** Focus retention campaigns on high-risk geographic areas, prioritize high-value customers with declining revenue, use customer segmentation for personalized marketing, and monitor key churn indicators proactively.
- **Validation:** AI-generated findings were validated using exploratory data analysis, machine learning results, and business logic to ensure consistency and reliability.

8. Final Conclusions

This project demonstrates the complete data analytics lifecycle, beginning with raw customer data and ending with actionable business insights.

Key achievements include:

- Successful preprocessing of telecom customer data.
- Identification of major churn drivers.
- Development and comparison of multiple predictive models.
- Selection of the best-performing model for churn prediction.
- Creation of an interactive Power BI dashboard for executive decision-making.
- Generation of AI-assisted insights to support customer retention strategies.

The project highlights how machine learning and business intelligence can work together to improve customer retention and support strategic business decisions.